Final Project

Poisson Equation: Simulation of a cooking hob

SI1336 Simulation and Modeling

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Abstract

The aim of this project is to solve the 2D Poisson equation for a rectangle geometry and Dirichlet boundary conditions.

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \rho(x, y) = 0$$

We will consider the case of calculating the stationary distributions of the temperature in a rectangle. Our idea is to use this equation to simulate the behaviour of an electric cooking hob by solving the equation applying three different iterative methods:

- Jacobi
- Gauss-Seidel
- Successive Over-Relaxation

A will be compared to each other and afterwards, one will be able to decided which is the best for our particular case.

Structural considerations

Here there are the consideration chosen to simulate the cooking hob. Firstly, the dimensions of the hob are L_x =32.5 cm, $L_y = 44.5\,cm$ with the following boundary conditions: $T(0, y) = 17°\text{C}$ $T(x, L_y) = 11.2°\text{C}$ $T(L_x, y) = 25.3°\text{C}$ and $T(x, 0) = 0.5°\text{C}$. And initially h=0.5 cm.

Related to the structure of the rectangle, we consider three burners:

- **1st burner** is a centred circumference at (8, 22.5) cm which follows the model:$\rho_1(x, y) = \rho_{1,0} e^{-(r-5)^2/0.3^2}$ where $r = \sqrt{(x-8)^2 + (y-22.5)^2}$ and $\rho_{1,0} = 10°\text{C}/cm^2$.
- **2nd burner i**s another centred circumference at (22, 10.5) cm which follows the model: $\rho_2(x, y) = \rho_{2,0} e^{-(r-4)^2/0.8^2}$ where $r = \sqrt{(x-22)^2 + (y-10.5)^2}$ and $\rho_{1,0} = 8°\text{C}/cm^2$.
- **3rd burner,** this is a bit different, we have considered a small rectangle of 4cmx6cm centred at (20, 32), and while inside the rectangle we have $\rho_3(x, y) = 3°\text{C}/cm^2$, outside, $\rho_3(x, y) = 0°\text{C}/cm^2$

Finally, we have: $\rho(x, y) = \rho_1(x, y) + \rho_2(x, y) + \rho_3(x, y)$

Mathematical considerations

To solve the Poisson equations, first of all, the equation has to be discretized:

$$T_{[i+1,j]} + T_{[i-1,j]} + T_{[i,j+1]} + T_{[i,j-1]} - 4_{T[i,j]} = -\rho_{(i,j)}$$

Some important parameters to take into account: ***k*** is the number of the iteration, we consider ***k*** as *old* value of the matrix, while ***k+1*** are the *new* values, ***i*** and ***j*** are the position of the coordinates, ***h*** is the grid spacing, as smaller value, as better precision. In addition, in the case of SOR, $\boldsymbol{\omega}$ is used and the best values are near 1.5.

As it has already been said, to solve the Poisson equation, three iterative methods will be used and compared. Here, there is a summary of each one:

<u>Jacobi method</u>

$$T_{[i,j]}^{k+1} = \frac{T_{[i+1,j]}^{k} + T_{[i-1,j]}^{k} + T_{[i,j+1]}^{k} + T_{[i,j-1]}^{k} + \rho_{(i,j)}h^2}{4}$$

<u>Gauss-Seidel</u>

$$T_{[i,j]}^{k+1} = \frac{T_{[i+1,j]}^{k} + T_{[i-1,j]}^{k+1} + T_{[i,j+1]}^{k} + T_{[i,j-1]}^{k+1} + \rho_{(i,j)}h^2}{4}$$

<u>Successive Over-Relaxation (SOR)</u>

This method is very similar to Gauss-Seidel, but a new parameter is included, $1 < \omega < 2$.

$$T_{[i,j]}^{k+1} = (1-\omega)T_{[i,j]}^{k} + \frac{\omega\left(T_{[i+1,j]}^{k} + T_{[i-1,j]}^{k+1} + T_{[i,j+1]}^{k} + T_{[i,j-1]}^{k+1} + \rho_{(i,j)}\, h^2\right)}{4}$$

Results

Comparison of three iterative methods

Here we have the results for the temperature calculated at one specific point, it has been chosen the (15.5, 23.5) cm, and the simulation has been run with the following parameters: $\mathrm{T_{in}} = 0°\mathrm{C}$, 10,000 iterations, h =0.5 cm, and ω=1.35.

In figure (1) are shown these results, where one can appreciate that the fastest iterative method to reach the final value is the Successive Over-Relaxation, followed by Jacobi and afterwards, Gauss-Seidel.

However, it is important to highlight that the precision and speed of Successive Over-Relaxation changes with the value of $\boldsymbol{\omega}$, as in figure (5).

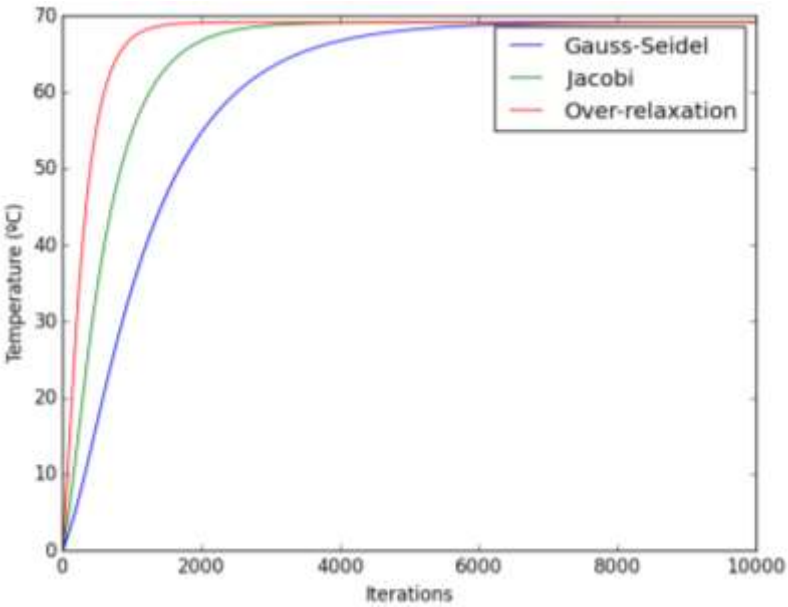

Figure 1: *Temperature vs iterations, where it can be seen that SOR is the fastest method to reach the final temperature (with* $T_{ini} = 0°\mathrm{C}$ *and* $\omega = 1.35$*)*

Dependence with the initial temperature

In this section, it has been studied the behaviour of the simulation changing the initial value of the temperature at the interior of the cooking hob. Looking at the following figures simulated with 10,000 iterations, h=0.5 cm and ω=1.35 and temperatures $\mathrm{T_{in}} = 0;\ 10;\ 120 \text{ and } 1040\ °\mathrm{C}$, one can conclude that the initial condition of the temperature is not relevant for the final result, as the four figures (1, 2, 3 ,4) have the same final temperature.

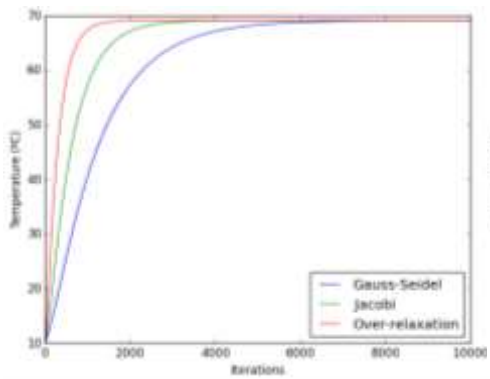

Figure 2*: Temperature vs iterations,* $T_{ini} = 10°\mathrm{C}$

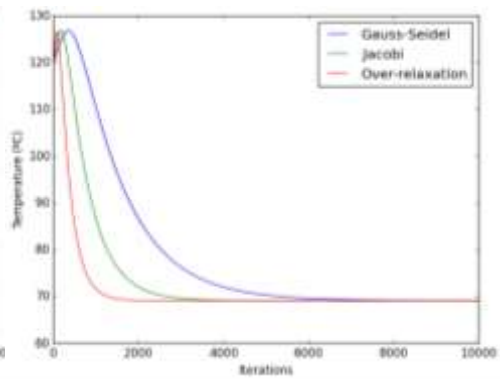

Figure 3: *Temperature vs iterations,* $T_{ini} = 120°\mathrm{C}$

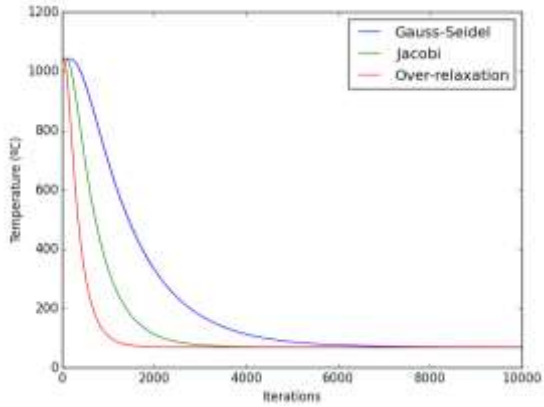

Figure 4: *Temperature vs iterations,* $T_{ini} = 1040°\mathrm{C}$

SOR method: Dependence with ω

In this section, one will focus on the effect of the parameter ω when we use the Sucessive Over-Relaxation method. To study this behaviour, the following parameters have been fixed, $T_{ini} = 10°\mathrm{C}, h = 0.5\ cm,\ 4{,}000$ iterations, and different values for $\omega = 1.2, 1.35, 1.5, 1.7$

In the this figure, it is appreciated that in this case, as higher is the value of ω, as faster is the SOR method.

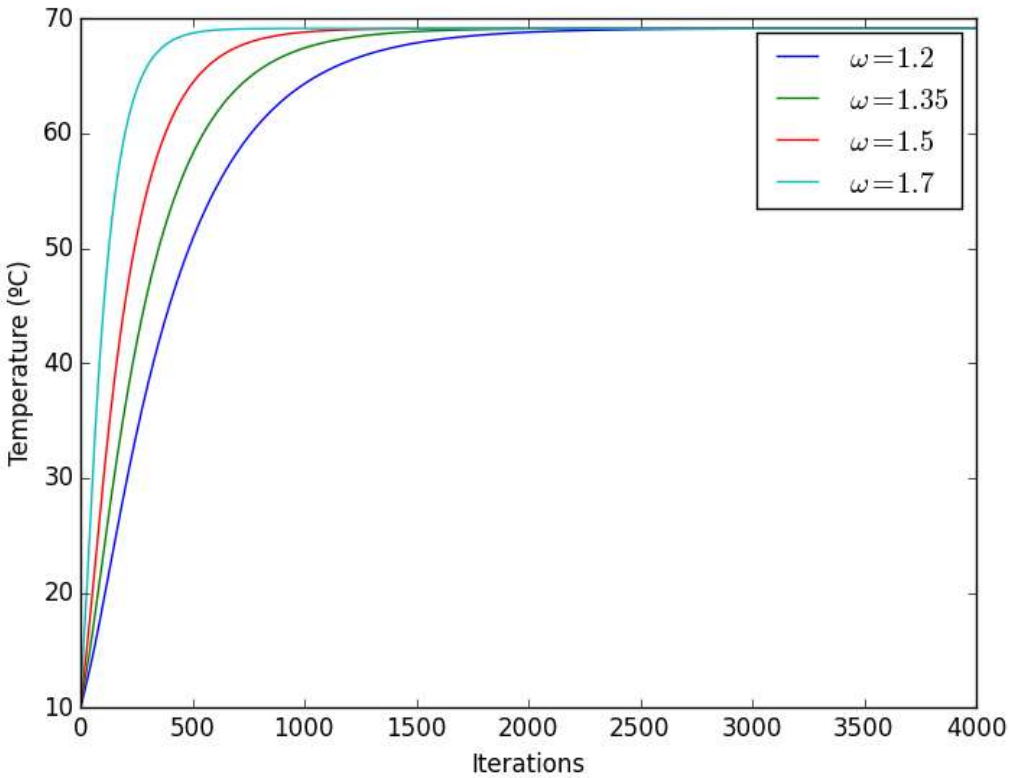

Figure 5: *Temperature vs iterations, studying the SOR method, and comparing different values of* $\boldsymbol{\omega}$*. The most efficient for this case is* $\boldsymbol{\omega} = \mathbf{1.7}$.

Going back to the section where the three methods were compared, it can be emphasised that the best more efficient method studied is the SOR with $\omega = 1.7$, it uses less iterations to reach the same final value.

The effect of *h*

Here, we want only to highlight that the size of parameter h is very import, as bigger is, less precision, but faster is the simulation, while if h is very small, we can obtain a high precision, but then, it takes much more time. For this reason, in this simulation it has always been consider h=0.5 cm as it is a balanced value.

Distribution of temperatures

At this last point, the distribution of temperatures on the rectangle which has been considered the cooking hob. On the first figure (6) it is shown the initial distribution, before any iteration just with the density of the three burners. While, on the second (7), there is the final temperature distribution, considering an initial temperature of 10°C.

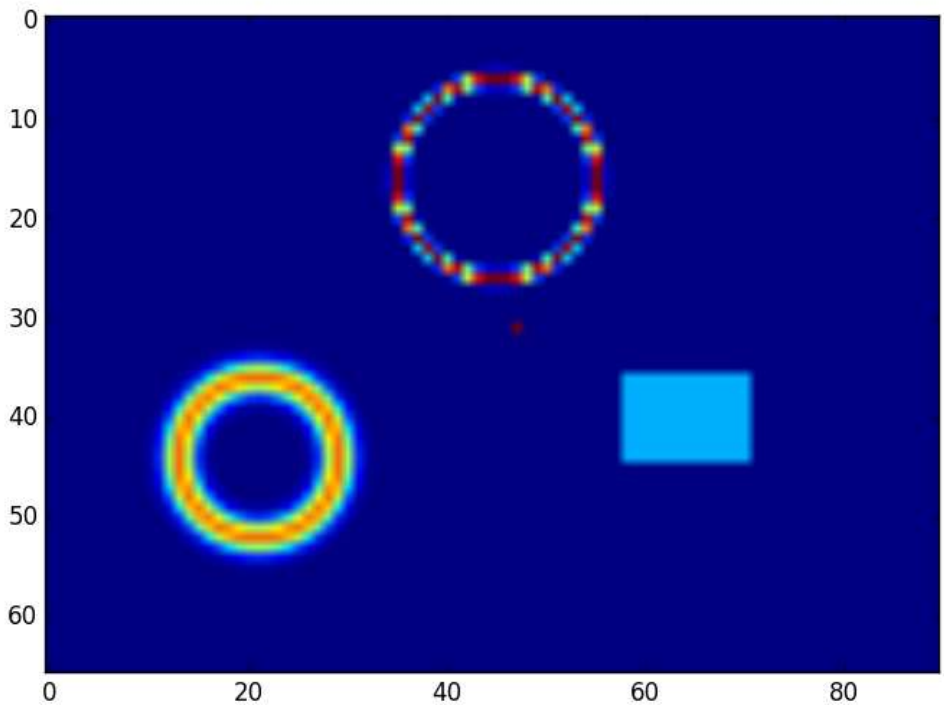

Figure 6: *Initial distribution of the density caused by the three burners. The red point in the middle is the one chosen to calculate the temperature in the previous figures.*

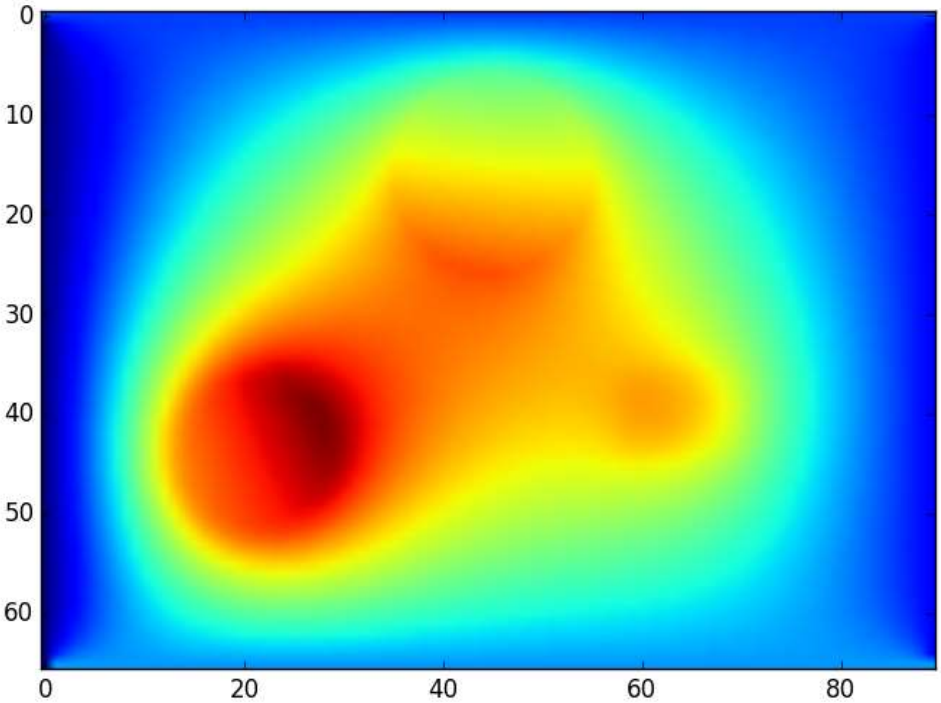

Figure 7: *Final distribution of the temperature in the cooking hob. It can be appreciated that the zones with the highest temperatures are red while the ones with the lowest are blue.*

Conclusion

All in all, with this project the Poisson equation has been solved following three iterative methods. According to our results, one can conclude that the most efficient method been used has been the SOR with ω=1.7, however, after a high number of iterations, all methods reached the same result.

Moreover, it has been seen that the initial condition of temperature does not affect the final result,